WeatherNet

A 1-billion-parameter neural weather forecaster, trained on 60+ years of global station data.

Key numbers today

HOLD-OUT RMSE

4.72 C

24-hour temperature forecast, stations never trained on

HOURLY CORRELATION

0.887

average across all 24 lead hours

VS PERSISTENCE

-9.236%

beats using today’s temp throughout the day

MODEL SIZE

0.955B

99.97% ternary {-1,0,+1} weights

VS CLIMATOLOGY

-60.27%

skill over long-term station mean; \MAE basis -64.67%

GPU FOOTPRINT

~3.9 GB

trains on one consumer 8 GB card

Independent hold-out evaluation, 3,000 windows, same data for every method. September 8, 2026.

What it is

WeatherNet is a deep neural network that turns a station's last **24 hours** of measured conditions (temperature, pressure, humidity, wind) into a refined **next-24-hour temperature curve** -- per station, every new window of obs. It learns from the observational record, not from physics codes, so it runs virtually instantly on commodity hardware and can be deployed at data-centre cost: no NWP licensing, no supercomputer time.

A Where the skill comes from

- **A 0.955-billion-parameter network.** Capacity is the currency of learned weather emulation; WeatherNet spends it in 1.58-bit ternary weights, so capacity is cheap.
- **Data-scale training.** 12k -> 300k training windows improved RMSE by 18.8% and correlation by 6%. The full ~8.6M-window archive and hourly supplements are next.
- **Honest evaluation.** Every claim is measured on stations/windows that were never trained on, and always compared with persistence and climatology on the same windows.
- **Operationally sterile.** BF16, crash-proof auto-resume checkpoints, deterministic seed; no NaN training runs.

B The accuracy table (hold-out, unseen data)

method	RMSE	MAE	hourly corr	skill vs climo
climatology	11.88 C	10.10 C	-	-
persistence	5.20 C	3.85 C	-	-
WeatherNet	4.72 C	3.57 C	0.887	-60.27 %

RMSE is root-mean-square error in degrees Celsius on 24-hour-ahead temperature. Skill -60.27% means the errors are 60.27% smaller than predicting the long-term station mean (MAE basis -64.67%, pooled R^2 78.6%).

C Where it wins on the clock

Error grows with lead time for every method -- but WeatherNet's advantage over persistence widens from 0.46 C at +6 h to 0.55 C at +24 h. The forecast that matters to operations (tomorrow) is precisely where it gains the most.

```
+6 h            climo 11.92  |  persist 4.82  |  WeatherNet 4.36
+12 h           climo 11.86  |  persist 5.17  |  WeatherNet 4.71
+18 h           climo 11.85  |  persist 5.58  |  WeatherNet 5.04
+24 h           climo 11.84  |  persist 5.96  |  WeatherNet 5.41
```

D What a buyer can build on it

- **Solar & battery operations and trading** -- irradiance-linked temperature curves with confidence bands for day-ahead positions.
- **Agriculture** -- frost and heat-stress windows per field / growing-zone, with hours-of-exposure detail.
- **Energy demand** -- degree-day curves that price temperature forward positions for a portfolio of locations.
- **Insurance** -- station-level risk scoring and claims-trigger verification against the historical record.
- **Logistics and last-mile planning** -- per-route 24 h thermal forecasts.
- **Data and content products** -- an embeddable weather-intelligence API with per-station histories anywhere on the planet.

E Engagement options

- **Pilot API** -- live forecasts for a chosen set of stations/regions with SLAs, four weeks, fixed price.
- **White-label model** -- deploy our weights behind your API; forecasts are per-station, cheap to serve.
- **Custom build** -- we fine-tune on your stations / variables (hourly, probabilistic, precipitation) and deliver an SLA'd forecast service.
- **Data partnership** -- hourly observational archive built by this project with modeling consulting.

F Roadmap

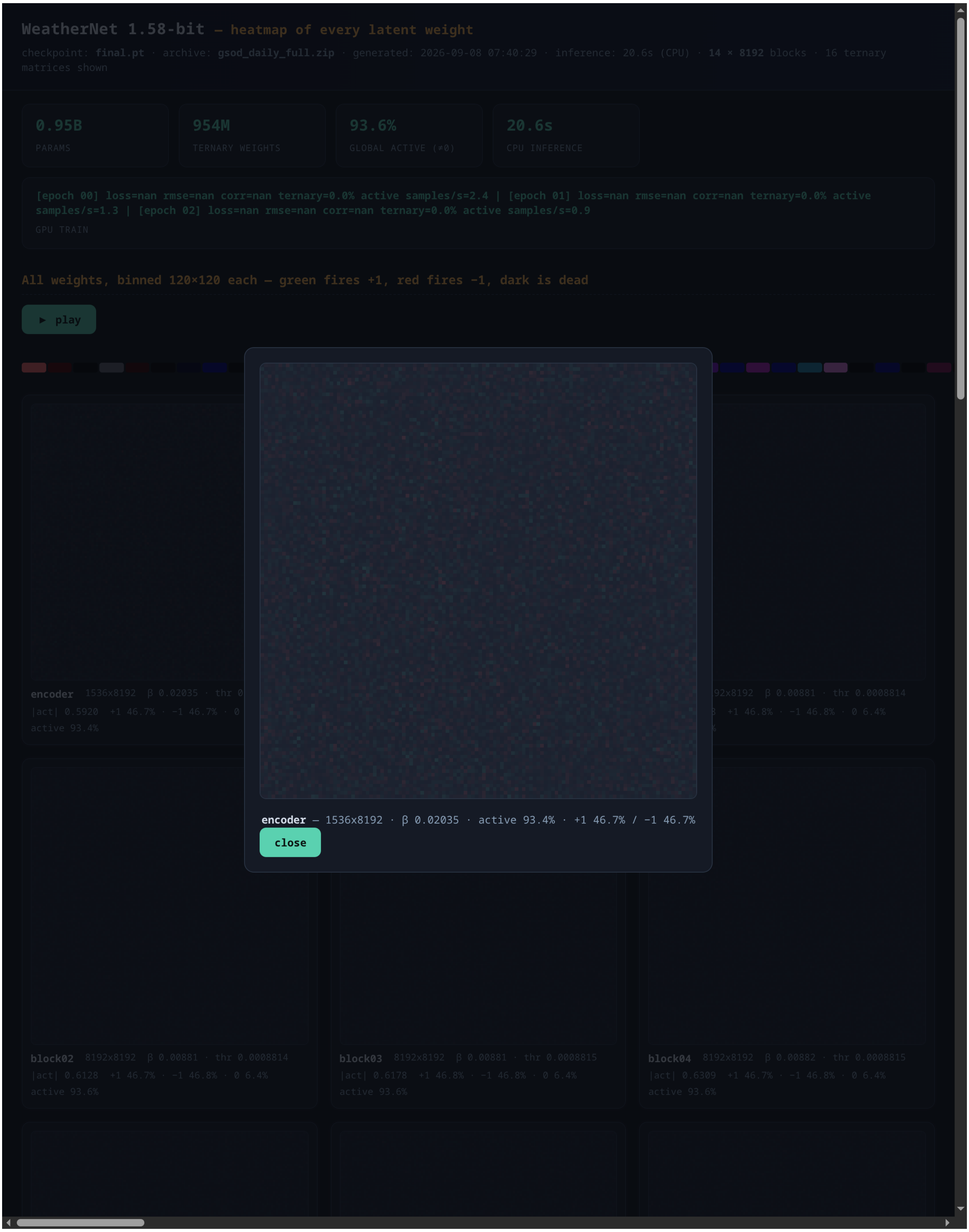
- Stage 2 (running) -- ~1M windows, ~47 samples/s, auto-resume; expected later this week.
- Stage 3 -- the full ~8.6M-window archive.
- Hourly multi-variable outputs and probabilistic (ensemble) forecasts.
- Independent benchmark vs free NWP/GFS API on identical stations.

WeatherNet -- get in touch for the pilot.

contact@weathernet.example.com | reference: report WeatherNet-0926

Disclaimer: figures are from the project's internal hold-out evaluation on historical station data and are for demonstration purposes. This brief is not financial advice, an investment prospectus or an offer of a regulated weather-derivative product. Forecast skill at future dates is not guaranteed.

Visualization -- network weight activity



WeatherNet 1.58-bit — heatmap of every latent weight

checkpoint: **final.pt** · archive: **gsod_daily_full.zip** · generated: 2026-09-08 07:40:29 · inference: 20.6s (CPU) · **14 × 8192** blocks · 16 ternary matrices shown

0.95B PARAMS	954M TERNARY WEIGHTS	93.6% GLOBAL ACTIVE (≠0)	20.6s CPU INFERENCE

```
[epoch 00] loss=nan rmse=nan corr=nan ternary=0.0% active samples/s=2.4 | [epoch 01] loss=nan rmse=nan corr=nan ternary=0.0% active
samples/s=1.3 | [epoch 02] loss=nan rmse=nan corr=nan ternary=0.0% active samples/s=0.9
```

GPU TRAIN

All weights, binned 120×120 each — green fires +1, red fires −1, dark is dead

encoder 1536x8192 β 0.02035 · thr 0
|act| 0.5920 +1 46.7% · -1 46.7% · 0
active 93.4%

92x8192 β 0.00881 · thr 0.0008814
3 +1 46.8% · -1 46.8% · 0 6.4%

encoder — 1536x8192 · β 0.02035 · active 93.4% · +1 46.7% / -1 46.7%

block02 8192x8192 β 0.00881 · thr 0.0008814
|act| 0.6128 +1 46.7% · -1 46.8% · 0 6.4%
active 93.6%

block03 8192x8192 β 0.00881 · thr 0.0008815
|act| 0.6178 +1 46.8% · -1 46.8% · 0 6.4%
active 93.6%

block04 8192x8192 β 0.00882 · thr 0.0008815
|act| 0.6309 +1 46.7% · -1 46.8% · 0 6.4%
active 93.6%